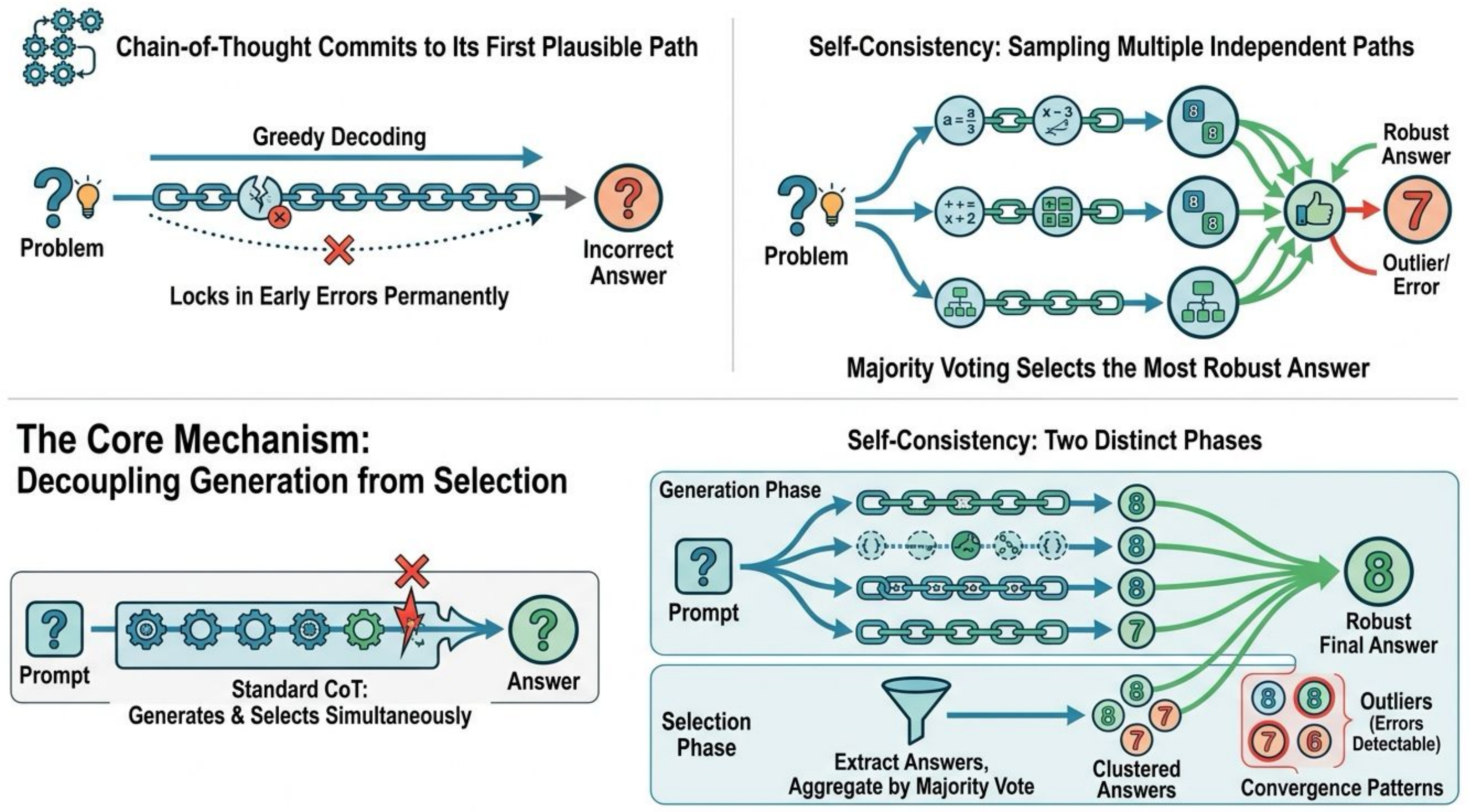
U.S. General Services Administration

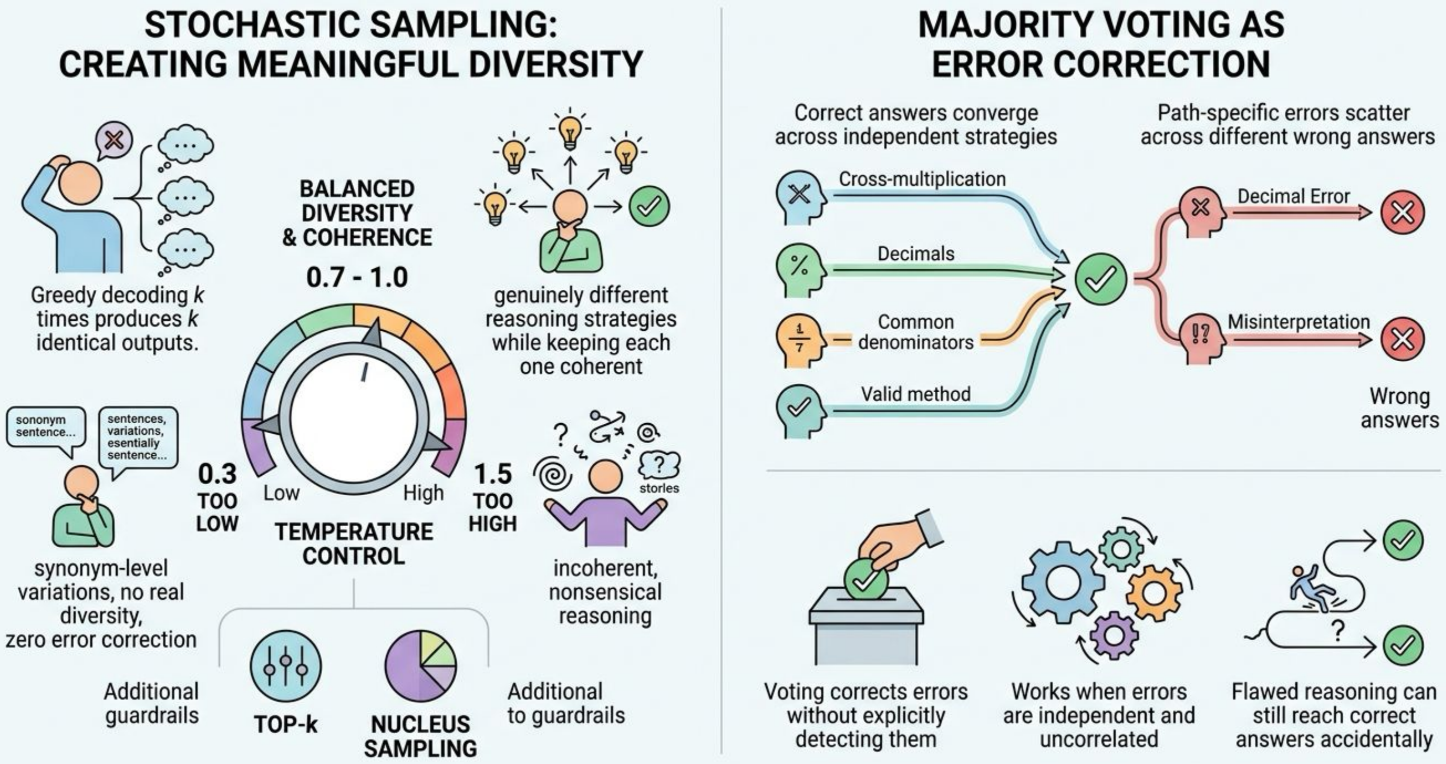
Chapter 5.3: Self-Consistency Fundamentals

US General Service Administration
AI Community of Practice - March 2026

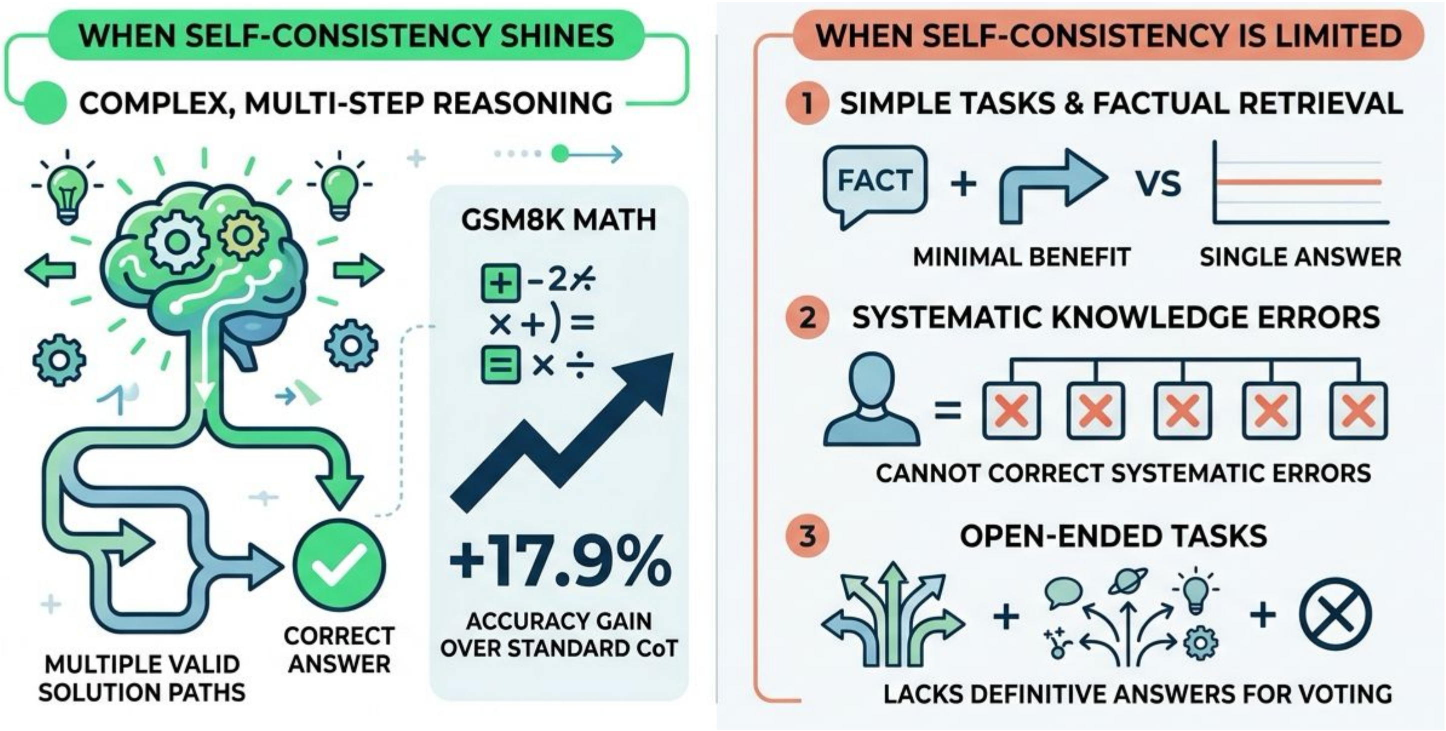
The Core Mechanism



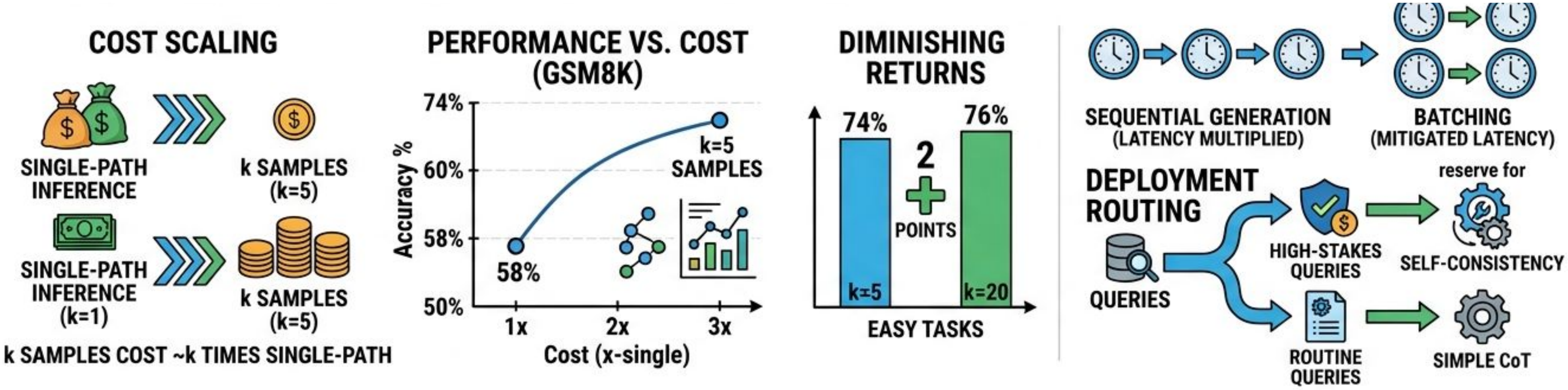
Stochastic Sampling & Majority Voting



When Self-Consistency Shines -- and Not Shine

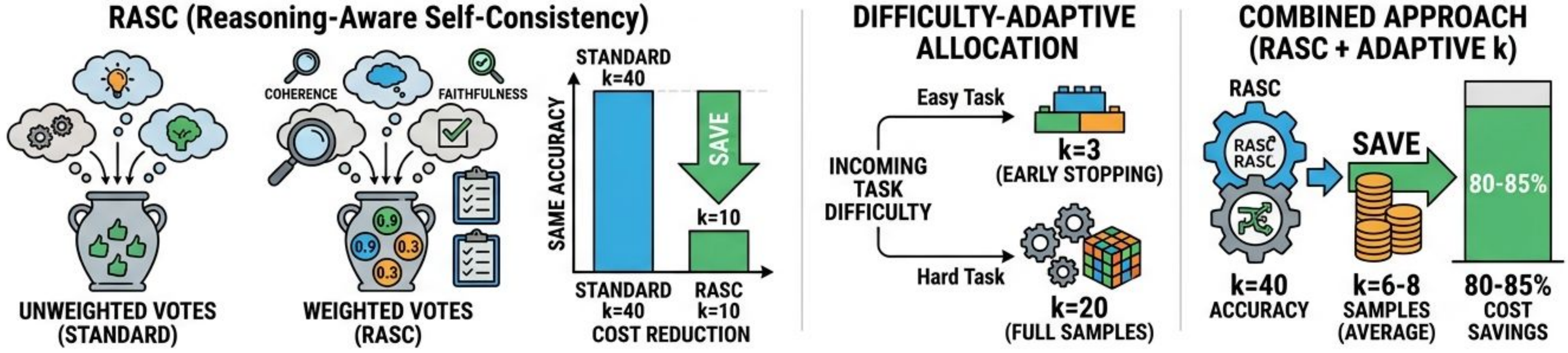


Cost-Accuracy Trade-offs in Production



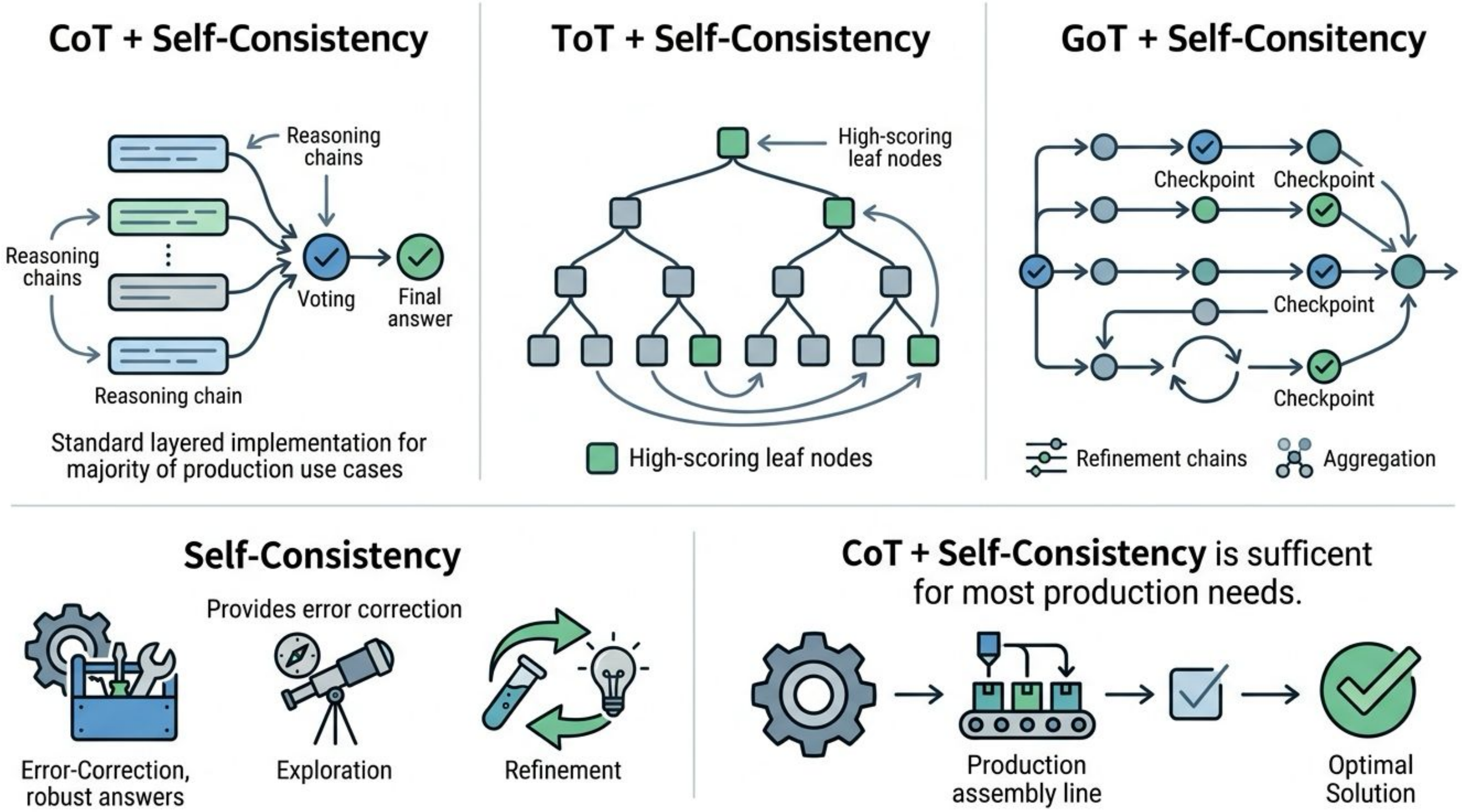
- k samples cost approximately k times single-path inference
- k=5 on GSM8K: 58% to 74% accuracy at 3x cost
- Diminishing returns: k=5 to k=20 adds 2 points on easy tasks
- Latency multiplies for sequential generation (mitigated by batching)
- Reserve Self-Consistency for high-stakes queries

Smarter Sampling -- RASC and Adaptive Allocation

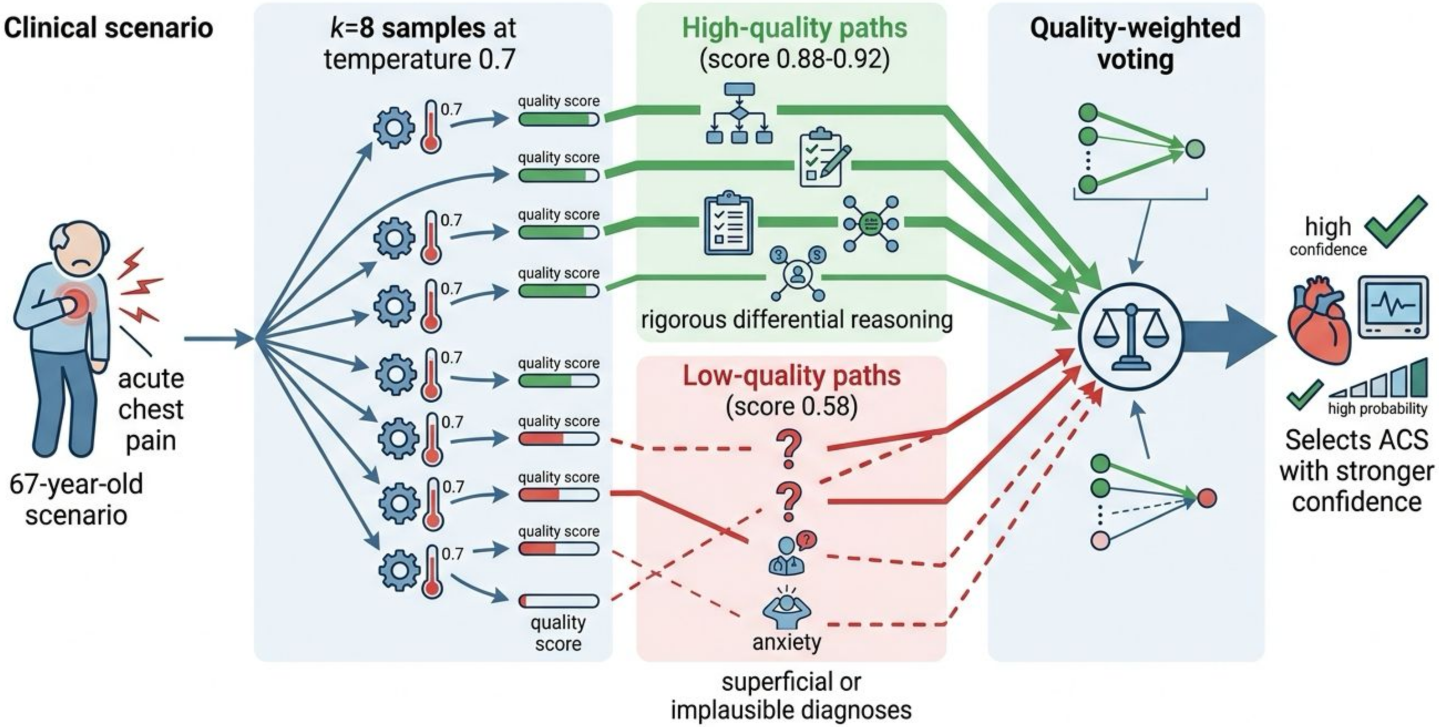


- Not all reasoning paths deserve equal votes
- RASC weights votes by coherence, faithfulness, relevance
- Achieves k=40 accuracy with only k=10 samples (70% cost reduction)
- Difficulty-adaptive allocation: k=3 for easy, k=20 for hard
- Combined approach averages k=6-8 for 80-85% cost savings

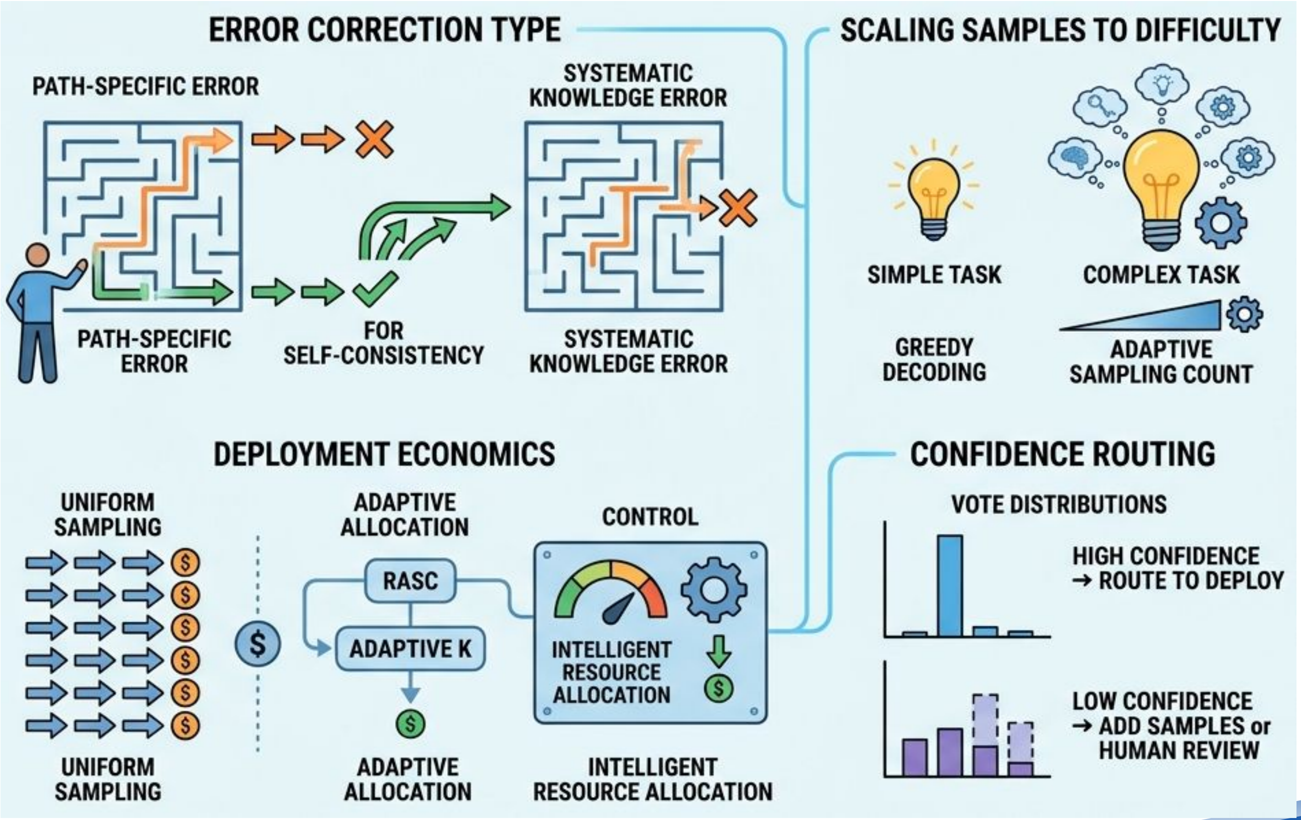
Integration with CoT, ToT, and GoT



Case: Medical Diagnosis with Weighted Voting



Optimizing Self-consistency



Key Takeaways

- Self-Consistency corrects path-specific errors, not systematic ones
- Match sample count to problem difficulty and error economics
- RASC and adaptive k make production deployment economical
- Vote distributions enable intelligent confidence-based routing
- Next: Chapter 5.4 tackles hierarchical planning for multi-step tasks